Ask and Ask Again: Repeated Screening for Smoking Increases Likelihood of Prescription for Cessation Treatment Among Women Veterans

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ABSTRACT

BACKGROUND: Preventive screening at the point of care can increase desired clinical outcomes. However, the impact of repeated screening for tobacco use on receiving smoking cessation treatment among women Veteran population has not been documented.

OBJECTIVE: To examine screening for tobacco use using clinical reminders and the association between the number of screenings and prescription for cessation treatment.

DESIGN: A retrospective analysis using data from a 5-year implementation trial for cardiovascular risk identification conducted between December 2016 and March 2020.

SUBJECTS: Women patients who had at least one primary care visit with a women's health provider during the study period at five primary care clinics in the Veterans Affairs (VA) Healthcare System.

MEASURES: The outcome is prescription of pharmacotherapy or referral to behavioral counseling for smoking cessation on or after the screening date. The exposure is the number of screenings for tobacco use from the trial and the annual VA national clinical reminders during the study period.

RESULTS: Of 6009 eligible patients, 5788 (96.3%) were screened at least once for tobacco use over five calendar years, and 2784 of those screened (48.1%) were reported as current and former smokers. Among current and former smokers, 709 (25.5%) received a prescription and/or referral for smoking cessation. In the adjusted model, the average predicted probability of prescription and/or referral for smoking cessation was 13.7% among current and former smokers screened once over 5 years, 18.6% among screened twice, 26.5% among screened thrice, 32.9% among screened four times, and 41.7% among screened five or six times.

CONCLUSIONS: Repeated screening was associated with higher predicted probabilities of being prescribed smoking cessation treatment.

Keywords: Tobacco; Smoking cessation; Preventive services; Primary care; Veterans

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INTRODUCTION

Tobacco use is a major cause of cardiovascular disease and the leading cause of preventable death in the USA.[1] The US Preventive Services Task Force recommends the Five A's approach to screen all patients at the point of care regardless of risk factors and counsel them to quit smoking: (1) Ask patients about current tobacco use; (2) Advise them to quit; (3) Assess their willingness to quit; (4) Assist them with counseling, medication, or resources to quit; and (5) Arrange follow-up with patients.[2] The 2008 Task Force highlights evidence that implementation of a clinical reminder system increases rates of screening ("Ask"), cessation treatment ("Assist"), and treatment follow-up ("Arrange").[3] In 1999, the Department of Veterans Affairs (VA) healthcare system adopted a practice to screen all VA patients for tobacco use annually regardless of smoking status[4] through an integrated clinical reminder within the VA electronic health record (EHR). Upon a positive screen for tobacco use, VA recommends that providers advise the patient to quit and offer evidence-based treatment options to support cessation attempts.[5]

National smoking rates have decreased in recent years[6]; however, certain vulnerable populations continue to be at increased risk for the deleterious effects of tobacco use. In particular, the national prevalence of smoking among

women Veterans (19.1%) is higher than among men Veterans (16.2%) and civilian women (13.1%).[7] On average, women Veterans, especially younger Veterans, have a greater burden of mental health conditions (e.g., posttraumatic stress disorder, depression, and military sexual trauma) that are known to increase the likelihood of smoking and make quitting difficult.[8] In addition, about one in four women Veterans are diagnosed with hypertension, diabetes, high cholesterol, or overweight/obese status that can elevate cardiovascular risk,[9] and smoking elevates the risk of cardiovascular conditions.[1,10–12] Therefore, it is critical to support smoking cessation among women Veterans of all ages by screening and offering evidence-based treatments for smoking cessation.

Smoking cessation has been found to be most effective with behavioral counseling and the use of medications such as nicotine replacement therapy (NRT), bupropion sustained-release, or varenicline.[13] However, prior studies on the use of these services among women Veterans were either done over a decade ago[14,15] or on a limited sample size.[16] An updated understanding of the smoking status and cessation efforts among women Veterans is needed.

Integration of an automated smoker identification system such as annual clinical reminders in the VA EHR represents an opportunity to screen smoking and assess rates of delivering cessation treatment among women Veterans. The current study examines the frequency of screening by the annual reminder in VA EHR among women Veterans and its association with rates of referrals for smoking cessation pharmacotherapy and/or behavioral counseling while adjusting for patient demographics and physical and mental comorbidities associated with increased cardiovascular risk.

METHODS

Participants

The study utilized the cohort from an implementation trial, Cardiovascular Risk Screening and Risk Reduction in Women Veterans (CV Toolkit; ClinicalTrials.gov NCT02991534), which was part of the larger implementation effort in VA: Enhancing Mental and Physical health of Women through Engagement and Retention (EMPOWER) Quality Enhancement Research Initiative (QUERI; QUE 15–272) program.[17] The CV Toolkit was implemented at five VA primary care clinics within three VA Healthcare Systems that were part of the VA Women's Health Practice-Based Research Network.[18] Utilizing a non-randomized stepped wedge design, the CV Toolkit was implemented via three components: (1) a patient self-administered screener to increase identification of personal CV risk; (2) a provider template in the VA EHR designed to enhance patient-provider communication and shared decision-making about CV risk; and (3) a facilitated group to provide a supportive, coordinated health coaching intervention to facilitate women's goal-setting and engagement and retention in appropriate health services.[19] Both the patient screener and provider template included a smoking screener. All women patients regardless of Veteran status were included if they had a VA primary care visit with a women's health provider between December 1, 2016, and March 31, 2020, at one of the five sites. The Department of Veterans Affairs Central Institutional Review Board approved the study.

Data Sources

We performed a retrospective analysis using data from the VA Corporate Data Warehouse (CDW), a national repository of the EHR covering VA clinical and administrative systems. Datasets used in the study included referrals for services, outpatient and inpatient care services, pharmacy records, patient clinical information and demographics, and national clinical reminders as well as the CV provider template which were captured in the Health Factor dataset.

Outcome: Prescription for Smoking Cessation Treatment

The outcome was a measure of physician orders for smoking cessation treatment, which included provider prescriptions for medications and/or referrals to behavioral counseling identified on or after the date of tobacco use screening. Prescription orders delivered by the VA Pharmacy services were searched in the outpatient prescription dataset: "nicotine", "bupropion 150 mg", "varenicline 0.5 mg", or "varenicline 1 mg". Referrals to behavioral counseling were identified in the consult dataset by searching for the following text fields in the provider requested services: "tobacco", "smoking", "nicotine", or "quit". We built a dichotomous outcome variable coded "yes" for receipt of a prescription for medications and/or referral to behavioral counseling (else "no").

Exposure: Screening for Tobacco Use

Tobacco use was defined as the use of any of the following: cigarettes, cigars, pipe smoking, snuff, dip, or chewing tobacco in the VA clinical reminder, and did not include electronic cigarettes, vaping devices, or other electronic nicotine delivery systems. Screening for tobacco use was administered by VA providers using annual VA clinical reminders and the CV Toolkit provider template.

Using a previously validated algorithm based on VA data,[5] we defined the following smoking status categories from the health factor values from the VA national clinical reminder. Current smoker included text fields such as "current tobacco user", "tobacco dependence", or "active smoker." Former smokers included patients who were not current users at any point over the study period and had "tobacco quit", "former tobacco user", or similar text fields. Non-smokers were patients who were not current or former smokers at any point

and had "lifetime non-user of tobacco", "never smoked", or similar fields. For the CV Toolkit provider template, "currently smoke or use tobacco some/everyday" responses were coded as current smoker, "smoked or used tobacco" and "currently do not smoke or use tobacco some/everyday" were coded as former smokers, and "not smoked or used tobacco" were non-smokers. Patients who did not have any of the above health factor values were considered not screened. We categorized current smokers if patients were ever screened as current smokers at any point during the search timeframe, former smokers if ever screened as former smokers but not as current smokers during the timeframe, and non-smokers if screened as non-smokers throughout the timeframe.

We calculated annual tobacco screenings received per patient over 5 years (December 1, 2016–March 31, 2020). A patient could receive up to five screenings if screened annually by the VA national reminder. As part of the study, a patient could receive a one-time CV Toolkit provider template in addition to the national reminder. As such, any patient could be screened up to six times during the five calendar-year study period (range 1–5, with 5 and 6 collapsed since only ten patients were screened six times).

Covariates

Patient characteristics included age and race/ethnicity recorded at the time of the first primary care visit during the study period. We included physical comorbidities that are linked with increased risk of cardiovascular disease: hypertension, high cholesterol, overweight/obesity, diabetes, prediabetes, renal failure, abdominal aortic aneurysm, stroke, acute myocardial infarction, and cardiac arrest (see Appendix Table 1 for ICD-9 and 10 codes). Each physical condition was identified based on the presence of relevant diagnosis codes on two records in the outpatient setting or one record in the inpatient setting during the two years prior to their first primary care visit in the study period.[20] The total number of physical comorbidities was calculated per patient and grouped into a count indicator (0, 1, 2, 3, 4, or 5 +) for analyses. Mental health comorbidities including posttraumatic stress disorder (PTSD) and depression were identified using the same criteria as physical conditions (Appendix Table 1). We also identified military sexual trauma (MST) in medical records, which was coded as positive if patients answered "yes" to any of the two questions: "While you were in the military 1) Did you receive uninvited and unwanted sexual attention, such as touching, cornering, pressure for sexual favors, or verbal remarks? 2) Did someone ever use force or threat of force to have sexual contact with you against your will?" We included Veteran status and service-connected disability that predict the level of access to VA care. Service-connected disability indicates whether the Veterans Benefits Administration (VBA) identifies and recognizes a medical condition or diagnosis as related to the patient's military experience. A higher rating of service-connected disability (range 0–100%) means a higher level of compensation by VBA and access to a broader and more comprehensive range of VA healthcare services available to the Veteran.[21]

Analyses

We assessed differences in patient characteristics between current and former smokers using the chi-squared test for categorical variables and *t*-test for continuous variables. Patients who were never screened in the study period ($N = 221/6009$, 3.7%) and non-smokers ($N = 3004/6009$, 50.0%) were excluded from analyses.

We used logistic regression to assess the relationship between the number of screenings and receipt of prescriptions for cessation treatment among current and former smokers and among current smokers only, controlling for patient characteristics and clustering by clinic. The number of screenings and the type of reminder (VA national clinical reminder vs. CV Toolkit) were not included together in regressions due to collinearity. We excluded number of primary care visits over the study period from regression due to a high correlation with screening frequency (correlation coefficient = 0.7885, $p < 0.05$). We computed the adjusted predicted probabilities of receiving a prescription by the number of screenings.[22,23] We conducted two separate sensitivity analyses — one excluding bupropion because it is also used for depression treatment and one including the screening via the CV Toolkit without the number of screenings. The study used Stata 15.1 for all analyses (StataCorp, College Station, TX).

RESULTS

Among 6009 eligible women patients, 96.3% ($N = 5788$) were screened for tobacco use at least one time over 5 years. Among those screened, 25.0% ($N = 1449$) were identified as current smokers and 23.1% ($N = 1335$) as former smokers. Among current and former smokers, 15.7% ($N = 438$) were screened once, 25.8% ($N = 717$) twice, 28.9% ($N = 804$) three times, 20.5% ($N = 571$) four times, and 9.1% ($N = 254$) five or six times.

Compared to former smokers, current smokers were younger and were more likely to be African American. While current smokers had a lower rate of physical comorbidities than former smokers, they were more likely to have PTSD, depression, self-reported MST, or a high service-connected disability ($p < 0.0001$ for each comparison; Table 1).

Overall, 25.5% of current and former smokers received a prescription for cessation treatment; 363 (13.0%) received NRT, 340 (12.2%) bupropion, 95 (3.4%) varenicline, and 128 (4.6%) referral to a behavioral counseling program. Table 2 further describes prescription rates by the frequency of screening overall and by current and former smokers.

In the fully adjusted logistic model, each additional screening was associated with higher odds of receiving a

Table 1 Patient Characteristics

Patient characteristics, *N* (%)	Overall (*N* = 2784)	Current smokers (*N* = 1449)	Former Smokers (*N* = 1335)	*p* value*
Age, mean (SD)	47.3 (14.7)	45.9 (14.0)	48.7 (15.3)	< 0.001
Race/ethnicity				
White	1171 (42.1)	579 (40.0)	592 (44.3)	< 0.001
Hispanic/Latino	421 (15.1)	198 (13.7)	223 (16.7)	
African American	791 (28.4)	474 (32.7)	317 (23.7)	
American Indian	22 (0.8)	13 (0.9)	9 (0.7)	
Asian	66 (2.37)	35 (2.4)	31 (2.3)	
Native Hawaiian	33 (1.2)	20 (1.4)	13 (1.0)	
Unknown/missing	280 (10.1)	130 (9.0)	150 (11.2)	
Count of physical comorbidities ‡				
0	1276 (45.8)	692 (47.8)	584 (43.8)	0.027
1	648 (23.3)	340 (23.5)	308 (23.1)	
2	466 (16.7)	231 (15.9)	235 (17.6)	
3	247 (8.9)	108 (7.5)	139 (10.4)	
4	122 (4.4)	68 (4.7)	54 (4.0)	
5+	25 (0.9)	10 (0.7)	15 (1.1)	
PTSD	959 (34.5)	580 (40.0)	379 (28.4)	< 0.001
Depression	1123 (40.3)	631 (43.6)	492 (36.9)	< 0.001
Military sexual trauma (MST)				
Reported MST	1138 (40.9)	649 (44.8)	489 (36.6)	< 0.001
Not reported MST	1523 (54.7)	736 (50.8)	787 (59.0)	
Unknown/declined	123 (4.4)	64 (4.4)	59 (4.4)	
Service-connected disability				
Non-service connected	726 (26.1)	342 (23.6)	384 (28.8)	< 0.001
0–49%	408 (14.7)	202 (13.9)	206 (15.4)	
50–99%	1091 (39.2)	560 (38.6)	531 (39.8)	
100%	559 (20.1)	345 (23.8)	214 (16.0)	
Veteran	2670 (95.9)	1386 (96.2)	1284 (95.7)	0.483

*Pearson chi-squared test for categorical variables and ANOVA for continuous variables

‡The count of physical conditions included diagnoses recorded in two outpatient visits or one inpatient visit in the last 2 years prior to the first primary care visit during the study period for hypertension, high cholesterol, overweight/obesity, diabetes, pre-diabetes, renal failure, abdominal aortic aneurysm, stroke, acute myocardial infarction, and cardiac arrest

Table 2 Receipt of Prescription for Tobacco Cessation Medication or Behavioral Counseling, Overall and By Current and Former Smokers

	Overall (*N* = 2784)	Current Smokers (*N* = 1449)	Former Smokers (*N* = 1335)
Received any prescription for treatment	709/2784 (25.5)	561/1449 (38.7)	148/1335 (11.1)
Type of treatment[a]			
Nicotine Replacement	363/2784 (13.0)	352/1449 (24.3)	11/1335 (0.8)
Bupropion 150 mg	340/2784 (12.2)	202/1449 (12.9)	138/1335 (10.3)
Varenicline 0.5 or 1 mg	95/2784 (3.4)	95/1449 (6.6)	0
Behavioral counseling	128/2784 (4.6)	128/1449 (8.8)	0
Number of tobacco use screenings			
1	87/438 (19.8)	35/197 (17.8)	18/241 (7.5)
2	131/717 (18.2)	102/361 (28.3)	29/356 (8.2)
3	212/804 (26.3)	170/427 (39.8)	42/377 (11.1)
4	199/571 (34.9)	159/315 (50.5)	40/256 (15.6)
5 or 6	114/254 (44.9)	95/149 (63.8)	19/105 (18.1)

[a]The type of treatment counted unique patients who received each type of treatment on or after the date of tobacco use screening

prescription (Table 3). Current smokers had 4.8 times the odds of getting a prescription compared to former smokers (95%CI, 2.4–9.8). Having PTSD or depression was associated with higher adjusted odds of receiving a prescription (1.2 [95%CI, 1.0–1.4] and 1.8 [95%CI, 1.3–2.5], respectively). Hispanic/Latino and African American patients had lower adjusted odds of receiving a prescription compared to White patients (0.9 [95%CI, 0.9–0.9] and 0.9 [95%CI, 0.9–1.0], respectively). However, older age (> 65), higher physical comorbidities, MST, or higher service-connected disabilities were not associated with differential odds of receiving a prescription. The model including current smokers only showed similar adjusted odds ratios of receiving prescriptions by the number of screening compared to the full model.

Predicted probabilities of receiving a prescription of pharmacotherapy and/or counseling among current and former smokers were 13.7% (95%CI, 7.9–19.5%) when screened once, 18.6% (95%CI, 14.3–22.9%) when screened twice, 26.5% (95%CI, 26.0–27.0%) when screened trice, 32.9% (95%CI, 25.3–40.4%) when screened four times, and 41.7% (95%CI, 35.2–48.3%) when screened five/six times (Fig. 1). Predicted probabilities among current smokers also increased with an additional screening.

Sensitivity analysis excluding bupropion (Appendix Table 2) showed similar adjusted odds ratios of receiving prescriptions by frequency of screening compared to the full model. However, depression was no longer significant in the model without bupropion. Sensitivity analysis without the number of screenings showed that being screened with the national reminder and a CV Toolkit template was associated with 1.7 times higher adjusted odds of receiving a prescription compared to being screened only with the national reminder (95%CI, 1.2–2.6) (Appendix Table 3). The predicted probability was higher among those with the CV Toolkit than without (31.8% [95%CI, 17.6–27.5%] vs. 22.6% [95%CI, 22.3–41.1%].

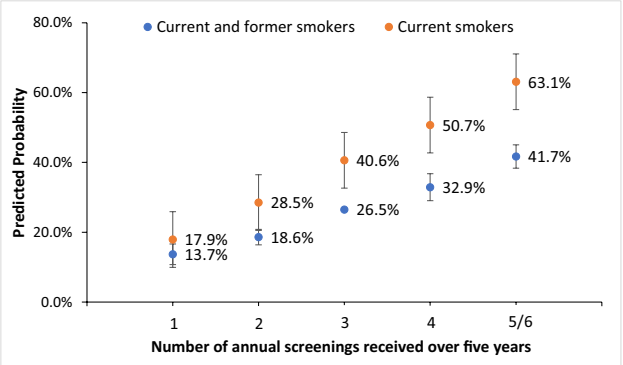

Fig. 1 Predicted probabilities of receiving any prescription for tobacco cessation medication and/or referral to behavioral counseling, by the number of annual tobacco use screenings over 5 years. Legend: The model controlled for age, race/ethnicity, current vs. former smoker, physical comorbidities, PTSD, depression, military sexual trauma, and service-connected disability. The 95% confidence interval for group 3 among current and former smokers is between 26.0% and 27.0%.

DISCUSSION

Screening for tobacco use with an annual clinical reminder significantly increased the predicted probability of receiving prescriptions for smoking cessation treatment among women Veteran current and former smokers, and the probability increased with each additional annual screen. Additional

Table 3 Adjusted Odds Ratio of Receiving any Prescription for Tobacco Cessation Medication And/or Referral to Behavioral Counseling

	Model among current + former smokers ($N = 2784$)	**Model among current smokers only ($N = 1449$)**
Independent variables	Odds Ratios (95% confidence interval)	Odds Ratios (95% confidence Interval)
Number of annual tobacco use screening (Ref: 1)		
2	1.5 (1.2–1.9) ***	1.8 (1.8–1.9) ***
3	2.5 (1.6–4.0) ***	3.2 (2.5–4.1) ***
4	3.6 (3.4–3.9) ***	4.9 (3.3–7.2) ***
5–6	5.7 (5.1–6.4) ***	8.2 (6.4–10.5) ***
Current smoker (Ref: former smoker)	4.8 (2.5–9.4) ***	N/A
PTSD	1.2 (1.0–1.4) *	1.3 (1.0–1.6) *
Depression	1.8 (1.3–2.5) *	1.2 (0.8–1.9)
Race/ethnicity (Ref: White)		
Hispanic/Latino	0.9 (0.9–0.9) **	0.8 (0.7–0.9) *
African American	0.9 (0.9–1.0) *	1.0 (0.9–1.3)
Others	0.9 (0.7–1.4)	0.8 (0.4–1.9)
Missing/unknown	0.9 (0.8–1.2)	1.0 (0.9–1.2)

***$p < 0.0001$; **$p < 0.01$; *$p < 0.05$

The models also controlled for age, physical comorbidities, military sexual trauma, and service-connected disability (each not statistically significant)

screening with a targeted reminder, such as the CV Toolkit, further increased the predicted probability of receiving a prescription. Our findings suggest that repeated annual screening is an important component in cessation intervention among women Veterans.

Our study is the first study to examine the use of automated clinical reminders in the VA EHR for assessing smoking status ("Ask") and treatment provision ("Assist") among women Veteran VA users. Ninety-six percent of women Veteran VA primary care users were screened over the 5-year period. The screening frequency was highly correlated with primary care visits, suggesting that systematic screening using the VA national screener mostly occurred at a primary care visit. We found that women Veterans who were repeatedly screened every year per VA recommendation had a statistically higher predicted probability of receiving a prescription for cessation therapy from providers than women with less than annual screens. Clinical practice guidelines recommend routine screening of smoking at least once a year in primary care followed by brief advice to quit, assessment of their readiness to quit, and offering cessation support. Patients who are prepared to quit and accept cessation support can then be offered cessation medications and behavioral counseling.[24–28] Our findings support the efficacy of the core objective of the guidelines to routinely ask all patients about their smoking status in offering cessation treatment.

Overall, 25% of current and former smokers received prescriptions and/or referrals to counseling for cessation treatment. This rate was higher than 16% of women Veteran smokers who received NRT in an earlier study using patient surveys,[14] yet lower than evidence in the 2008 PHS guideline and another study among Veterans.[15] One possible explanation is that our study examined patients with a prescription to quit smoking and referrals to cessation programs, whereas prior studies also included brief counseling to quit.[29] Another possible explanation is that our study focused on women, who in prior studies had a lower rate of being prescribed quit-smoking therapy than men.[14] Future research should focus on longitudinal follow-up to examine how the 5As delivered through clinical reminders ultimately translate into quit rates and whether there is a gender difference in receiving treatment with the use of reminders.

Smoking is a relapsing condition that requires multiple interventions and attempts to quit.[30] The risk of relapse can occur from about 50% among recent quitters who quit one year ago to about 10% who quit 10 years ago.[31,32] Recognizing that relapse is part of the process of cessation, ongoing support and care to decrease episodes of relapse is required.[33] Therefore, we included former smokers as potential relapsed smokers and examined how often they received cessation assistance. Eleven percent of former smokers received prescriptions possibly to prevent them from relapsing and their probability of receiving treatment increased with additional annual screening. Therefore, regular and repeated screening may benefit former smokers reduce relapse episodes.

Disparities observed in our study are consistent with other studies. Higher rates of smoking were observed among women Veterans who were younger or had PTSD or depression diagnoses.[34,35] Women patients who were Hispanic or Black had fewer prescriptions for tobacco cessation medications than White patients. Patients who had depression and PTSD were more likely to have a prescription for cessation medications than those without these conditions.[35] In sensitivity analysis removing bupropion from our model, depression was no longer significantly distinguished between those who received a prescription and who did not, while PTSD remained significant. These findings suggest that women Veterans with depression may be prescribed bupropion more frequently than other cessation medication, possibly given its dual indication for both depression and tobacco use.

Our study has a few limitations. First, the VA EHR did not capture the measure of a patient's readiness to quit, which is an important predictor of receiving cessation treatment. Second, clinical reminders during face-to-face interaction with patients may produce an underreport of smoking status. Third, while smoking patients may be asked about motivation and/or readiness to quit or smoking relapse between individual visits, this communication is not captured in clinical reminders, but rather documented in clinical notes. Therefore, we may have underestimated how often women Veterans were offered cessation medication or counseling. Fourth, routine tobacco use screening occurred mostly in primary care as part of VA recommendation. Due to this high correlation between primary care visits and screening, we were unable to account for the number of primary care visits and screenings together in the models. Bias could arise if patients were asked about smoking between the annual screenings which would not be captured as routine screening in EHR or when smokers had a higher number of primary care encounters in a year which may give more opportunities to have counseling to quit. However, despite potential underreporting, we still found that annual screening at a minimum impacted prescription rates for cessation.

Smoking cessation assistance is increased with repeated asking. Repeated annual screening combined with an intervention tool focused on cardiovascular risk discussions facilitated more smoking cessation treatment referrals with prescriptions. From these data, we recommend all primary care settings consistently ask patients at least annually about their smoking status and consider incorporating any innovations that may highlight an additional smoking discussion to enhance treatment acceptance.

Supplementary Information *The online version contains supplementary material available at https://doi.org/10.1007/s11606-023-08227-y.*

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Declarations

Conflict of Interest *The authors declare that they do not have a conflict of interest.*

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